

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	0.16
Total 1 mark			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates $P(X \leq 2)$ or $P(X \leq 3)$ using the binomial dist	AO3.4	M1	$P(X \leq 2) = 0.55(177\dots)$ $P(X \leq 3) = 0.80(589\dots)$
	Obtains correct answer AWRT 0.448	AO1.1b	A1	$P(X \geq 3) = 1 - P(X \leq 2) = 0.448$
(b)	Calculates the cube of their answer to (a). Do not accept as part of a larger calculation or multiples of their cube.	AO1.1a	M1	$0.448\dots^3$ $= 0.090$ (3dp)
	Obtains correct answer (FT if AWWF 0.0899 to 0.0901)	AO1.1b	A1F	
(c)	States first appropriate assumption in context	AO3.5b	E1	The probability of hitting the bullseye is fixed at 0.3 or
	States second appropriate assumption in context Accept probability of hitting bullseye 'is constant' Do not accept 'fixed number of trials'	AO3.5b	E1	Hitting the bullseye with each dart is independent of hitting the bullseye with any other dart or There are 2 possible outcomes, hit bullseye or does not hit bullseye
				Total 6 marks

Q3.

Marking Instructions	AO	Marks	Typical Solution
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Circles correct answer	AO1.1b	B1	$\frac{1}{10}$
Total 1 mark			

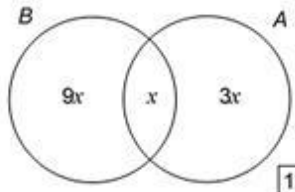
Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates $P(\text{studies Physics}) \times P(\text{studies Geography})$ or Calculates $P(\text{studies Geography} \text{studies Physics})$ or $P(\text{studies Physics} \text{studies Geography})$	AO3.1b	M1	$P(P) \times P(G) = \frac{12}{24} \times \frac{8}{24} = \frac{1}{6}$
	Shows $P(\text{studies Physics}) \times P(\text{studies Geography}) = P(\text{studies Physics} \cap \text{studies Geography})$ and correctly concludes that the events are independent or Shows that the appropriate conditional probability is equal to $P(\text{studies Geography})$ or $P(\text{studies Physics})$ and correctly concludes that the events are independent	AO2.1	R1	$P(P \cap G) = \frac{4}{24} = \frac{1}{6}$ Hence $P(P) \times P(G) = P(P \cap G)$ Therefore events are independent
(b)	Uses conditional probability to calculate $P(M \cap B)$	AO3.1b	M1	$P(M \cap B) = P(M) \times P(B M)$
	Obtains the correct value of $P(M \cap B)$	AO1.1b	A1	$= \frac{1}{5} \times \frac{3}{8} = \frac{3}{40}$
	Uses the addition rule to calculate $P(M \cup B)$	AO1.1a	M1	$P(M) + P(B) - P(M \cap B)$
	Obtains the correct value of $P(M \cup B)$	AO1.1b	A1	$= \frac{1}{5} + \frac{1}{6} - \frac{3}{40}$ $= \frac{7}{24}$
				Total 6 marks

Q5.

Marking Instructions	AO	Marks	Typical Solution
Finds P(Drop and Beanstalk and Giant)	AO1.1a	M1	$\frac{80}{225} \times \frac{75}{224} \times \frac{70}{223}$
Multiplies by 6 to obtain correct answer	AO1.1b	A1	$\frac{80}{225} \times \frac{75}{224} \times \frac{70}{223} \times 6 = 0.224$
Total 2 marks			

Q6.

Marking Instructions	AO	Marks	Typical Solution
Uses conditional probability, either ① or ②	AO3.1b	M1	$\frac{P(A \cap B)}{P(A)} = \frac{1}{4}$ ① $\Rightarrow P(A) = 4P(A \cap B)$
Obtains both equations ① and ② correctly	AO1.1b	A1	$\frac{P(A \cap B)}{P(B)} = \frac{1}{10}$ ② $\Rightarrow P(B) = 10P(A \cap B)$
Evaluates $P(A \cup B)$ correctly	AO1.1b	B1	$P(A \cup B) = 1 - \frac{122}{200} = \frac{39}{100}$
Uses addition law	AO1.1a	M1	$P(A) + P(B) - P(A \cap B) = \frac{39}{100}$ ③
Combines the three equations	AO1.1a	M1	$4P(A \cap B) + 10P(A \cap B) - P(A \cap B) = \frac{39}{100}$
Obtains correct probability, as a fraction or decimal	AO2.2b	A1	$P(A \cap B) = \frac{3}{100}$
ALT Produces a relevant Venn diagram	AO3.1b	M1	OR 200 
Labels Venn diagram	AO1.1b	A1	122

	correctly			
	Forms correct equation to find x PI	AO1.1b	B1	$9x + x + 3x = 200 - 122$
	Combines terms	AO1.1a	M1	$13x = 78$
	Solves equation	AO1.1a	M1	$x = 6$
	Obtains correct probability	AO2.2b	A1	$P(A \cap B) = \frac{6}{200}$ or 0.03
Total 6 marks				

Q7.

(a) (i) $P(F \& C) = \underline{0.3 \text{ or } 3 / 10 \text{ or } 30\%}$

Ratios (eg 3:10) are only penalised by 1 accuracy mark at first correct answer

CAO (0.3)

B1 (1)

(ii) $P(G \text{ or } S) = \underline{0.45 \text{ or } 45 / 100 \text{ or } 45\%}$

CAO (0.45)

B1 (1)

(iii) $P(C | F) = \frac{0.3 \text{ or (i)}}{0.55} =$

30 / 55 or 6 / 11

CAO (6 / 11)

M1

or

(0.54 to 0.55) or (54% to 55%)

AWFW (0.54545)

A1 (2)

(iv) $P(R' | D) = \frac{0.25 \text{ or } (0.30 - 0.05)}{0.30}$

Correct numerator

M1

Correct denominator

M1

25 / 30 or 5 / 6

CAO (5 / 6)

or

(0.83 to 0.834) or (83% to 83.4%)

AWFW (0.83333)

A1

(3)

(v) $P(F | C') = \frac{0.25 \text{ or } (0.60 - 0.35)}{0.60}$
Correct expression

M1

25 / 60 or 5 / 12

CAO (5 / 12)

or

(0.416 to 0.42) or (41.6% to 42%)

AWRT (0.41667)

A1

(2,3)

9

(b) $P = [P(F \& C)]^2 + [P(F \& G)]^2$

*Attempt at **sum of at least 2 squared***

terms; 0 < term < 1; not (a + b)²

*May be implied by a **correct** expression or a **correct** answer*

M1

$0.30^2 + 0.25^2$ or $0.09 + 0.0625 =$

OE

Ignore additional terms or integer multipliers

*May be implied by a **correct** answer*

A1

1525 / 10000 or 305 / 2000 or 61 / 400

CAO

or

(0.1525)

(0.152 to 0.153) or (15.2% to 15.3%)

AWFW

A1

3

[12]

Q8.

(a) (i) $P(A = 2) = 0.90 \times 0.95 = \underline{0.85 \text{ to } 0.86}$
AWFW (0.855 or 171 / 200 OE)

B1

(ii) $P(A = 1) = (0.90 \times 0.05) + (0.10 \times 0.95)$
*May be implied by a **correct** answer*

or $= 1 - [0.855 + (0.10 \times 0.05)]$
Do not ignore extra terms

M1

= 0.14
CAO (7 / 50 OE)

A1

3

(b) (i) $P(A_w \cap D_w) = 0.90 \times 0.80$
May be implied by a **correct** answer

M1

= 0.72
CAO (18 / 25 OE)

A1

2

$$(ii) \quad P(A_B \cap D_B) = (b)(i) \times 0.95 (\times 1)$$

$$\text{or} = 0.90 \times 0.80 \times 0.95 (\times 1)$$

or= (a)(i) × 0.80

May be implied by a **correct** answer

M1

0.68 to 0.685

AWFW (0.684 or 171 / 250 OE)

A1

2

$$(iii) \quad P(A_T \cap D'_T) = 0.95 \times 0 =$$

0

CAO; award on value only

B1

1

(iv) $P(\text{neither}) = P([A'_w \cap D'_w] \cap [A'_T \cap D'_T])$

$(1 - 0.90) \times (1 - 0.15)$

Accept 0.085 or 17 / 200 OE

Award M1 and m1 on value(s) only

M1

$(1 - 0.95) \times (1 - 0)$

Accept 0.05 or 1 / 20 OE

m1

or

$P(\text{neither}) =$

$P(A'_w \cap A'_T) \cap P(D'_w | A'_w) \cap P(D'_T | A'_T)$

$(1 - 0.90) \times (1 - 0.95)$

Accept 0.005 or 1 / 200 OE

Award M1 and m1 on value(s) only

(M1)

$(1 - 0.15) \times (1 - 0)$

Accept 0.85 or 17 / 20 OE

(m1)

$= 0.085 \times 0.05$ **or** 0.005×0.85
OE

= 0.0042 to 0.0043

AWFW

(0.00425 or 17 / 4000 OE)

A1

3

[11]

Q9.

(a)	A	B		
			——— E(0.15)	0.135
			——— T(0.9)	0.675
			——— L(0.10)	0.090
			——— T(0.35)	0.035
			——— L(0.1)	
			——— L(0.65)	0.065

Correct shape

B1

Correct labels

B1

Correct probabilities

B1

3

(b) (i) $P(E \cup T @ B) = 0.9 \times 0.9 + 0.1 \times 0.35$
 $1 - (0.09 + 0.065)$

M1

$= \underline{\mathbf{0.84 \text{ to } 0.85}}$
AWFW (0.845)

A1

2

(ii) $P(T @ A | T @ B) =$

$$\frac{0.9 \times 0.75}{(0.9 \times 0.75 + 0.1 \times 0.35)}$$
 $P(A | B) \text{ used in (ii) or (iii)}$

M1

$a \div (a + b)$ with at least a correct

m1

$= \frac{0.675}{0.71} = \underline{\mathbf{0.95 \text{ to } 0.951}}$
AWFW (0.95070)

A1

3

(iii) $P(L @ A | L' @ B) = \frac{0.1 \times 0.35}{(i)}$
F on (i)

AF1

$= \frac{0.035}{0.845} = \underline{\mathbf{0.04 \text{ to } 0.042}}$
AWFW (0.04142)

A1

2

(c) $P((T @ A | L @ B) \cap (T' @ A | L @ B))$

$$\frac{0.9 \times 0.1}{1 - 0.845} \times \frac{0.1 \times 0.65}{1 - 0.845} \times 2$$

First expression (18 / 31)

M1

Second expression (13 / 31)

M1

$\times 2$

M1

= 0.486 to 0.49

AWFW (0.48699)

A1

4

[14]

Q10.

Ratios (eg 194:640) are only penalised by 1 accuracy mark at first correct answer

(a) (i) $P(B = 3) = \underline{194 / 640 \text{ or } 97 / 320 \text{ or } 0.303 \text{ or } 30.3\%}$
 CAO or AWRT (0.303125)

B1

1

(ii) $P(T \geq 2) = \frac{172 + 256 + 135}{640} \text{ or } 1 - \frac{77}{640} \text{ or } \frac{563}{640} = \underline{563 / 640}$
 CAO

M1

or (0.879 to 0.88) or (87.9% to 88%)
 AWFW (0.879688)

A1

2

(iii) $P(B = 3 \text{ \& } T > 2) = \frac{72 + 99 + 16}{640} \text{ or } \frac{194 - 7}{640} \text{ or } \frac{187}{640}$

M1

= 187 / 640 or 0.292 or 29.2%
 CAO or AWRT (0.292188)

A1

$$(iv) \quad P(B < 3 \mid T = 2) = \frac{(14+67+72)}{172} \text{ or } \frac{172-19}{172} \text{ or } \frac{153}{172}$$

Correct numerator
Correct denominator
(accept both $\div 640$)

M1
M1

$$= \underline{153 / 172}$$

CAO

A1

$$\text{or } \underline{(0.888 \text{ to } 0.89) \text{ or } (88.8\% \text{ to } 89\%)}$$

AWFW (0.889535)

$$(b) \quad P(2T \cap 3T \cap \geq 4T \mid B = 3) = \frac{72}{194} \times \frac{99}{193} \times \frac{16}{192}$$

Correct 3 values multiplied in numerator
Correct 3 values multiplied in denominator
0.371 $\times$ 0.513 $\times$ 0.083 (all AWRT)
 $\Rightarrow$ M1 M1 (OE products)

M1
M1

abc multiplied by 6 or 3
 $0 < (a, b \text{ \& } c) < 1$

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M1

$$= \underline{0.095 \text{ to } 0.0952}$$

AWFW (0.095187)

A1

- Notes**
- 1 Incorrect answer with no working $\Rightarrow$ 0 marks
 - 2 The **3 correct fractions / decimals** identified but not multiplied (eg added) $\Rightarrow$ M1 M0 M0 A0
 - 3 The **3 correct fractions / decimals** identified together with 0.016 (AWRT) $\Rightarrow$ M1 M1 M0 A0
 - 4 A denominator of ${}^{194}C_3 = 1198144 \Rightarrow$ M2 (2nd & 3rd M1 marks)

Q11.

**Ratios (eg 194:640) are only penalised by
1 accuracy mark at first correct answer**

(a) (i) $P(B = 3) = \underline{194 / 640 \text{ or } 97 / 320 \text{ or } 0.303 \text{ or } 30.3\%}$
 CAO or AWR T (0.303125)

B1

1

(ii) $P(T \geq 2) = \frac{172 + 256 + 135}{640} \text{ or } 1 - \frac{77}{640} \text{ or } \frac{563}{640} = \underline{563 / 640}$

M1

$= \underline{563 / 640}$
 CAO

$\text{or } \underline{(0.879 \text{ to } 0.88) \text{ or } (87.9\% \text{ to } 88\%)}$
 AFWW (0.879688)

A1

2

(iii) $P(B = 3 \text{ \& } T \geq 2) = \frac{72 + 99 + 16}{640} \text{ or } \frac{194 - 7}{640} \text{ or } \frac{187}{640}$

M1

$= \underline{187 / 640 \text{ or } 0.292 \text{ or } 29.2\%}$
 CAO or AWR T (0.292188)

A1

2

(iv) $P(B \leq 3 \mid T = 2) = \frac{(14 + 67 + 72)}{172} \text{ or } \frac{172 - 19}{172} \text{ or } \frac{153}{172}$

Correct denominator
 Correct numerator
 (accept both $\div 640$)

M1

M1

$= \underline{153 / 172}$
 CAO

$\text{or } \underline{(0.888 \text{ to } 0.89) \text{ or } (88.8\% \text{ to } 89\%)}$
 AFWW (0.889535)

A1

Answers as fractions, percentages or ratios lose
 accuracy (A & B) marks in (b) & (c)

- (b) (a)(i) $\times$ (a)(ii) $\neq$ (a)(iii)
Attempted

M1

since

$$0.303 \times 0.88 = \underline{0.265 \text{ to } 0.27 \neq 0.292}$$

AWFW & AWRT

A1

2

SC Any **correct fully-explained** reasoning, using other than answers from part (a), which results in an inequality ($\neq$) **with both sides as numerically correct decimals** (to 3 dp) $\Rightarrow$ B1 (eg $P(B = 3) = 0.303 \neq P(B = 3 | T = 2) = 72 / 172 = 0.419$) but no / unclear / incomplete reasoning or no / incorrect / incomplete numerical work $\Rightarrow$ B0

(c) $P(2T \cap 3T \cap \geq 4T | B = 3) = \frac{72}{194} \times \frac{99}{193} \times \frac{16}{192}$
Correct 3 values multiplied in numerator
Correct 3 values multiplied in denominator

M1

$$0.371 \times 0.513 \times 0.083 \text{ (all AWRT)}$$

$\Rightarrow$ M1 M1 (OE products)

M1

abc multiplied by 6 or 3

$$0 < (a, b \text{ \& } c) < 1$$

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$$= \underline{0.095 \text{ to } 0.0952}$$

AWFW

$$(0.095187)$$

A1

4

Notes 1 Incorrect answer with no working $\Rightarrow$ 0 marks

2 The **3 correct fractions / decimals** identified but not multiplied (eg added) $\Rightarrow$ M1 M0 M0 A0

3 The **3 correct fractions / decimals** identified together with 0.016 (AWRT) $\Rightarrow$ M1 M1 M0 A0

4 A denominator of ${}^{194}C_3 = 1198144 \Rightarrow$ M2 (2nd & 3rd M1 marks)

[14]

Q12.

(a) (i) $P(M \cap C) = \frac{175}{645} = \frac{35}{129} = 0.271$

AWRT; accept either correct fraction

B1

1

(ii) $P(M) = \frac{519}{645} = \frac{173}{215} = 0.804$ to 0.805

AWFW; accept either correct fraction

B1

1

(iii) $P(LD) = \frac{63}{645} = \frac{21}{215} = 0.097$ to 0.098

AWFW; accept either correct fraction

B1

1

(iv) $P(L|F) = \frac{94}{126} = \frac{47}{63}$
Accept $\frac{94}{645} \div \frac{126}{645}$

M1

= 0.746

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A1

AWRT

2

(b) $P(L \cap L|F) = \left(\frac{94}{126} \times \frac{93}{125} \right) \text{ or } \frac{8742}{15750}$
Or $\left(\frac{47}{63} \times \frac{93}{125} \right) \text{ or } \frac{4371}{7875} \text{ or } \frac{1457}{2625}$

B1

= 0.555

B1

AWRT

2

(c) $P(L \cap C \cap (LD + 0))$

$$= \frac{349}{645} \times \frac{193}{644} \times \frac{63 + 40}{643}$$

Correct numerator

M1

Correct denominator

M1

SC The three correct fractions identified
but not multiplied $\Rightarrow$ M1 M0 M0 A0

$\times 6$ or 3

Note that a denominator of $\left(\frac{645}{3} \right)$

M1

$= 0.155$ to 0.157

$\Rightarrow$ M2 (second and third M1 marks)
AWFW

A1

4

NB: 0.026 with no working $\Rightarrow$ M1 only
 $0.026 \times 6 = 0.156$ with no working $\Rightarrow$ 4 marks

[11]

Q13.

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(a) (i) $P(M \cap C) = \frac{175}{645} = \frac{35}{129} = 0.271$

AWRT; accept either correct fraction

B1

1

(ii) $P(M) = \frac{519}{645} = \frac{173}{215} = 0.804$ to 0.805

AWFW; accept either correct fraction

B1

1

(iii) $P(LD) = \frac{63}{645} = \frac{21}{215} = 0.097$ to 0.098

AWFW; accept either correct fraction

B1

1

$$(iv) \quad P(L|F) = \frac{94}{126} = \frac{47}{63}$$

$$\text{Accept } \frac{94}{645} \div \frac{126}{645}$$

M1

$$= 0.746$$

AWRT

A1

2

$$(v) \quad P(M|L') = \frac{519 - 255}{645 - 349} = \frac{175 + 54 + 35}{193 + 63 + 40}$$

Allow one arithmetic slip

M1

Allow one arithmetic slip

M1

$$= \frac{264}{296} = \frac{132}{148} = \frac{66}{74} = \frac{33}{37}$$

Any of these implies M1 M1

$$= 0.891 \text{ to } 0.893$$

AWFW

A1

3

$$(b) \quad P(L \cap L' | F) = \left(\frac{94}{126} \times \frac{93}{125} \right) \text{ or } \frac{8742}{15750}$$

$$\text{Or } \left(\frac{47}{63} \times \frac{93}{125} \right) \text{ or } \frac{4371}{7875} \text{ or } \frac{1457}{2625}$$

B1

$$= 0.555$$

AWRT

B1

2

(c) $P(L \cap C \cap (LD + O))$

$$= \frac{349}{645} \times \frac{193}{644} \times \frac{63+40}{643}$$

Correct numerator

M1

Correct denominator

M1

SC The three correct fractions identified
but not multiplied $\Rightarrow$ M1 M0 M0 A0

$\times 6$ or 3

*Note that a denominator of $\binom{645}{3}$
 $\Rightarrow$ M2 (second and third M1 marks)*

M1

$= 0.155$ to 0.157

AWFW

A1

4

NB: 0.026 with no working $\Rightarrow$ M1 only
 $0.026 \times 6 = 0.156$ with no working $\Rightarrow$ 4 marks

[14]

Q14.

(a) $R \sim (15, 0.45)$ ©2025 Exam Papers Practice. All rights reserved.

(i) $P(R \leq 5) = 0.26$ (0) to 0.261
AWFW (0.2608)

B1

1

(ii) $P(R > 10) = 1 - P(R \leq 10)$
*Requires '1 -'
Accept 3dp rounding or truncation*

$= 1 - (0.9745 \text{ or } 0.9231)$
*Can be implied by 0.025 to 0.026 but not
by 0.0769 to 0.077*

M1

$$= 0.025 \text{ to } 0.026$$

AWFW (0.0255)

A1

2

(iii) $P(R = 6) = 0.4522 - (a)(i) \text{ or } = \binom{15}{6} (0.45)^6 (0.55)^9$
Can be implied by a correct answer

M1

$$= 0.191 \text{ to } 0.192$$

AWFW (0.1914)

A1

2

(iv) $P(5 \leq R \leq 10) = 0.9745 \text{ or } 0.9231 (p_1)$
Accept 3dp rounding or truncation
 $p_2 - p_1 \Rightarrow M0 M0 A0$
 $(1 - p_2) - p_1 \Rightarrow M0 M0 A0$
 $p_1 - (1 - p_2) \Rightarrow M1 M0 A0$
only providing result > 0

M1

Minus 0.1204 or 0.2608(p_2)
Accept 3dp rounding or truncation

M1

$$= 0.853 \text{ to } 0.855$$

AWFW (0.8541)

A1

3

Or
 B (15, 0.45) terms stated for at least 3 values within
 $4 R \leq 11$ gives probability
Can be implied by a correct answer

(M1)

$$= 0.853 \text{ to } 0.855$$

AWFW (0.8541)

(A2)

(b) (i) $P(S) = 0.85 \text{ plus } 1 \text{ minus}$
CAO; requires 'plus' or 'minus'

B1

$(0.15 \times 0.80) \quad (0.15 \times 0.20)$
 CAO; not simply 0.12 or 0.03

B1

= 0.97
 AG

2

NB: $(0.85 \times 0.20) + 0.80 \Rightarrow$ B0 B0
 $(0.85 \times 0.20) + (0.85 \times 0.80) + (0.15 \times 0.80) \Rightarrow$ B0 B1

- (ii) $P(S \geq 48) = 0.81$ to 0.82 or 0.5553 or 0.9372
Accept 3dp rounding or truncation
M2 for the three correctly expressed terms
for B (50, 0.03) or B (50, 0.97) added

M2

= 0.81(0) to 0.811
 AFWW (0.8108)

A1

3

NB: Answer = 0.4447 or 0.1892 or 0.0628 $\Rightarrow$ M1 only

- (iii) $p = 1 - 0.85 = 0.15$
CAO; may be implied by correct answer
or correct expression for mean

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B1

Mean, $\mu = 80 \times 0.15 = 12$
 CAO

B1

2

SC Mean = 9.6 $\Rightarrow$ B1 only

[15]

Q15.

(a)

	<i>J</i>	<i>J'</i>	Total
<i>W</i>	0.55	0.10	0.65
<i>W'</i>	0.15	0.20	0.35

Total	0.70	0.30	1.00
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0.35 and 0.7; CAO

B1

0.55; CAO

B1

0.1 and 0.2; CAO

B1

Notes:

Use of Venn or tree diagrams **without** table completion $\Rightarrow$ B0 B0 B0

Printed table not completed but constructed and completed on a different page $\Rightarrow$ B1 B1 B1 max

Accept fractional answers

Do not accept percentages

3

(b) P(purchases exactly one)

$$= P(W \cap J') + 0.15$$

*Only c's equivalent to 0.10 **shown and added to 0.15***

*Can be implied by **correct answer***

M1

$$= 0.10 + 0.15$$

$$= \mathbf{0.25 \text{ or } 25/100 \text{ or } 5/20 \text{ or } 1/4}$$

CAO

A1

2

(c) (i) $P(W \cup J) = \mathbf{0.8} \neq P(W) + P(J) = \mathbf{1.35}$

or $P(W \cap J) = \mathbf{0.55} (>0)$; accept if indicated in a Venn diagram

or $P(W) + P(J) = \mathbf{1.35} > 1$ **or impossible**

*Any one of these three **seen***

Ignore contradictions, explanations & justifications

B1

(ii) $P(W | J) = 0.55/0.70 = \mathbf{0.79}$

$$\neq p(W) = \mathbf{0.65}$$

*Do **not** accept use of W' and/or J'*

AWRT

B1

or $P(J | W) = 0.55/0.65 = 0.85$

$\&\neq P(J) = 0.70$

or $P(W) \times P(J) = 0.45 \text{ to } 0.46$

$\&\neq P(W \cap J) = 0.55$

*Any one of these three **seen***

Ignore contradictions, explanations & justifications

AWFW

Bdep1

3

[8]

Q16.

(a) (i)

	<i>J</i>	<i>J'</i>	Total
<i>W</i>	0.55	0.10	0.65
<i>W'</i>	0.15	0.20	0.35
Total	0.70	0.30	1.00

0.35 and 0.7; CAO

B1

0.55; CAO

B1

0.1 and 0.2; CAO

B1

Notes:

Use of Venn or tree diagrams **without** table completion $\Rightarrow$ B0 B0 B0

Accept fractional answers

Do not accept percentages

Printed table not completed but constructed and completed on a different page $\Rightarrow$ B1 B1 B1 max

3

(ii) $P(\text{purchases exactly one})$

$= P(W \cap J') + 0.15$

$= 0.10 + 0.15$

*Only c's equivalent to 0.10 **shown and added to** 0.15*

*Can be implied by **correct** answer*

M1

= 0.25 or 25/100 or 5/20 or 1/4
CAO

A1

2

(iii) **(A) $P(W \cup J) = 0.8 \neq P(W) + P(J) = 1.35$**

or $P(W \cap J) = 0.55 (>0)$; accept if indicated in a Venn diagram

or $P(W) + P(J) = 1.35 > 1$ or impossible

Any one of these three seen

Ignore contradictions, explanations & justifications

B1

(B) $P(W | J) = 0.55/0.70 = 0.79$

*Do **not** accept use of W' and/or J'*

AWRT

B1

$\neq P(W) = 0.65$

or $P(J | W) = 0.55/0.65 = 0.85$

$\neq P(J) = 0.70$

or $P(W) \times P(J) = 0.45$ to 0.46

$\neq P(W \cap J) = 0.55$

Any one of these three seen

Ignore contradictions, explanations & justifications

AWFW

Bdep1

3

(b) Do not allow multiplying factors in (b)

(i) $P(0) = 0.15 \times 0.40 \times 0.45$

*Can be implied by **correct** answer or
 $1 - (0.2265 + 0.466 + 0.2805)$*

B1

= 0.027 or 27/1000

CAO

B1

2

**(ii) $P(2) = 0.85 \times 0.60 \times 0.45 = 0.2295$
 $+ 0.85 \times 0.40 \times 0.55 = 0.1870$**

$$+ 0.15 \times 0.60 \times 0.55 = 0.0495$$

For either method:

At least two bold expressions correct

Only one bold expression correct

Can be implied by **correct** answer

M2
(M1)

or

$$= 1 - (0.027 + 0.2265 + 0.2805)$$

For second method:

Must have '1 -' for any marks

$$= 0.466 \text{ or } 466/1000 \text{ or } 233/500$$

CAO; **do not** imply this from (i)

A1

3

[13]

Q17.

(a) (i) $P(G) = 0.15$
CAO

B1

1

(ii) $P(A \cap \leq 1) = 0.60 \times 0.55 = 0.33$
CAO

B1

1

(iii) $P(\leq 24) = (0.60 \times 0.80) + (0.25 \times 0.85) + (0.15 \times 0.75)$
May be implied

M1

$$= 0.48 + 0.2125 + 0.1125 = 0.805$$

CAO

A1

2

(iv) $P(B \mid \leq 24) = \frac{P(B \cap \leq 24)}{P(\leq 24)}$
Used; may be implied

M1

$$= \frac{0.25 \times 0.85}{(iii)} = \frac{0.2125}{0.805}$$

F on (iii)

AF1

= 0.264

AWRT

A1

3

(b) (i) $P(3 @ B | \leq 24) = [(a)(iv)]^3$

Used; may be implied

M1

= 0.018 to 0.0185

AWFW (0.01839)

A1

2

(ii) $P(\text{same station} | \leq 24) = [P(A | \leq 24)]^3 + (b)(i) + [P(G | \leq 24)]^3$

Used; may be implied

M1

$$= \left(\frac{0.48}{0.805} \right)^3 + (b)(i) + \left(\frac{0.1125}{0.805} \right)^3$$

At least 1 term correct; allow (b)(i) providing it is a (cond prob)³

M1

All 3 terms correct

M1

= 0.2120 + 0.0184 + 0.0027 = 0.233

AWRT (0.23312)

A1

4

[13]

Q18.

Ratios (eg 65:1000) are only penalised by 1 mark at first correct answer

(a) (i) $P(\text{all 3 walk}) = 0.65 \times 0.40 \times 0.25$

*Can be implied by **correct** answer*

M1

$$= 65/1000 = 13/200 = 0.065$$

CAO; do not confuse with 0.65

A1

2

(ii) $P(\text{Rita by bus}) = 0.25 \times (1 - 0.15) \times (1 - 0.20)$
*Can be implied by **correct** answer*

M1

$$= 17/100 = 0.17$$

CAO

A1

2

(iii) $P(2 \text{ cycle})$
 $= 0.10 \times 0.45 \times (0.25 + 0.20)$
 $= 0.02025$
 $+ 0.10 \times (0.40 + 0.15) \times 0.55$
 $= 0.03025$
 $+ (0.65 + 0.25) \times 0.45 \times 0.55$
 $= 0.22275$
 (0.27325)
CAO at least 1 of these 3 terms or equivalent but allow a '× 3'

B1

$$P(3 \text{ cycle}) = 0.10 \times 0.45 \times 0.55$$

$$= 0.02475$$

CAO

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$$P(\geq 2 \text{ cycle}) = P(2 \text{ cycle}) + P(3 \text{ cycle})$$

Sum of 4 or 7 terms each a product of 3 probabilities but not '× 3'

M1

$$= 0.298$$

CAO

A1

or

$$P(0 \text{ cycle}) = 0.90 \times 0.55 \times 0.45 =$$

$$0.22275$$

CAO

(B1)

$$\begin{aligned}
 &P(1 \text{ cycles}) \\
 &= 0.10 \times 0.55 \times 0.45 = 0.02475 \\
 &+ 0.90 \times 0.45 \times 0.45 = 0.18225 \\
 &(0.47925)
 \end{aligned}$$

CAO at least 1 of these 3 terms but allow a '× 3'

(B1)

$$\begin{aligned}
 &+ 0.90 \times 0.55 \times 0.55 = 0.27225 \\
 &P(\geq 2 \text{ cycle}) \\
 &= 1 - [P(0 \text{ cycle}) + P(1 \text{ cycles})] \\
 &1 - [\text{sum of 4 terms each a product of 3 probabilities but not '× 3'}]
 \end{aligned}$$

(M1)

$$1 - 0.702 = 0.298$$

CAO

(A1)

4

(b) (i) $P(WW) = (0.65 \times 0.90) = 0.585$
 $P(CC) = (0.10 \times 0.70) = 0.070$
 CAO either

B1

$$P(WW \text{ or } CC) = 0.585 + 0.070$$

Sum of 2 terms each a product of 2 probabilities

M1

$$= 0.655$$

CAO; or equivalent

A1

3

(ii) $P(\text{different}) = 1 - (b)(i) = 0.345$
F on (b)(i) providing $0 < p < 1$

B1F

1

[12]

Q19.

$$P(C) = 0.6 \quad P(C \cap B) = 0.25$$

$$\{P(C \text{ only}) = 0.35 \quad P(B \text{ only}) = 0.4\}$$

*In (a), ratios (eg 4 : 10) are only
penalised by 1 mark
at first correct answer*

(a) (i) $P(C') = 1 - P(C) = 1 - 0.6 = 0.4$

CAO; or equivalent

B1

1

(ii) $P(C \cap B') = 0.6 - 0.25$

$$= 1 - (0.4 + 0.25)$$

Can be implied by correct answer

M1

$$= 0.35$$

CAO; or equivalent

A1

2

(iii) $P(B) = (i) + p$ with $p < 0.6$

Can be implied by correct answer

M1

$$= (i) + 0.25$$

Can be implied by correct answer

A1

$$= 0.65$$

CAO; or equivalent

A1

OR

$$P(B) = 1 - (ii)$$

Can be implied by correct answer

(M2)

$$= 0.65$$

(A1)

OR

$$1 = P(C) + P(B) - P(C \cap B)$$

Can be implied by correct answer

(M1)

Thus $P(B) = 1 - (0.6 - 0.25)$
Can be implied by correct answer

(A1)

$= 0.65$
CAO; or equivalent

(A1)

3

(b) $P(L | G_C) = 0.9$ $P(L | G_{CB}) = 0.7$
 $P(L | G_B) = 0.3$

$P(G \cap L) \Rightarrow (a)(ii) \times 0.9$ (0.315)
Follow through or correct

M1

0.25×0.7 (0.175)

M1

$[(a)(iii) - 0.25] \times 0.3$ (0.12)
Follow through or correct

M1

Note: Each pair of multiplied probabilities must be > 0 to score the corresponding method mark

Ignore any multiplying factors
Ignore any additional terms

$\Rightarrow 0.315 + 0.175 + 0.12 = 0.61$

CAO

A1

4

[10]

Q20.

$P(C) = 0.6$ $P(C \cap B) = 0.25$
 $\{P(C \text{ only}) = 0.35 \quad P(B \text{ only}) = 0.4\}$

In (a), ratios (eg 4: 10) are only penalised by 1 mark at first correct answer

(a) (i) $P(C') = 1 - P(C) = 1 - 0.6 = 0.4$
CAO; or equivalent

B1

(ii) $P(C \cap B') = 0.6 - 0.25$

Can be implied by correct answer

M1

$$= 1 - (0.4 + 0.25)$$

$$= 0.35$$

CAO; or equivalent

A1

(iii) $P(B) = (i) + p$ with $p < 0.6$

Can be implied by correct answer

M1

$$= (i) + 0.25$$

Can be implied by correct answer

A1

$$= 0.65$$

CAO; or equivalent

A1

OR

$$P(B) = 1 - (ii)$$

Can be implied by correct answer

EXAM PAPERS PRACTICE (M2)

$$= 0.65$$

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(A1)

OR

$$1 = P(C) + P(B) - P(C \cap B)$$

Can be implied by correct answer

(M1)

$$\text{Thus } P(B) = 1 - (0.6 - 0.25)$$

Can be implied by correct answer

(A1)

$$= 0.65$$

CAO; or equivalent

(A1)

3

$$(b) \quad P(L | G_C) = 0.9 \quad P(L | G_{CB}) = 0.7 \\ P(L | G_B) = 0.3$$

$$(i) \quad P(G \cap L) \Rightarrow (a)(ii) \times 0.9 \quad (0.315) \\ \text{Follow through or correct}$$

M1

$$0.25 \times 0.7 \quad (0.175)$$

M1

$$[(a)(iii) - 0.25] \times 0.3 \quad (0.12) \\ \text{Follow through or correct}$$

M1

Note: Each pair of multiplied probabilities must be > 0 to score the corresponding method mark

Ignore any multiplying factors
Ignore any additional terms

$$\Rightarrow 0.315 + 0.175 + 0.12 = 0.61 \\ \text{CAO}$$

A1

4

$$(ii) \quad \text{Probability} = \{1 - (b)(i)\}^5 \\ \text{Allow } 5 \times \{1 - (b)(i)\}^5$$

M1

$$= 0.39^5 = 0.009$$

AWRT (0.00902)

A1

2

[12]

Q21.

(a)

In (a), ratios (eg 100:160) are only penalised by 1 mark at first correct answer

$$(i) \quad P(P) = 100/160 = 50/80 = 25/40 = 10/16 = 5/8 = 0.625 \\ \text{CAO}$$

B1

1

$$(ii) \quad P(S') = 1 - \frac{32}{160} \quad \text{or} \quad P(S') = \frac{32}{160}$$

Or equivalent

Ignore labels of S' & S

*Can be implied by **correct** answer*

M1

$$= 128/160 = 64/80 = 32/40 = 16/20 = 8/10 = 4/5 = 0.8$$

CAO

A1

2

$$(iii) \quad P(S \text{ or } H) = P(S \cup H) = \frac{60 + 32 - 18}{160} \quad \text{or} \quad \frac{60 + 14}{160} \quad \text{or} \quad \frac{32 + 8 + 16 + 18}{160}$$

Or equivalent

*Can be implied by **correct** answer*

M1

$$= 74/160 = 37/80 = 0.462 \text{ to } 0.463$$

CAO/AWFW (0.4625)

A1

2

$$(iv) \quad P(T | P) = \frac{\frac{30}{160}}{(i)}$$

Or equivalent

*Can be implied by **correct** answer*

but watch for $\frac{18}{160}$ or $\frac{48}{160}$

M1

$$= 3/100 = 3/10 = 0.3$$

CAO

A1

2

$$(b) \quad P(1C \text{ \& } 1R \text{ \& } 1S) = \frac{24}{160} \times \frac{56}{159} \times \frac{32}{158}$$

Multiplication of any 3 different given subject totals

M1

Multiplication of 160, 159 & 158

M1

$$(0.15 \times 0.35220 \times 0.20253)$$

$\times 6$

Accept 3 dp accuracy

Award for $3 \leq \text{multiplier} \leq 6$

M1

$$= 0.064 \text{ to } 0.0644$$

AWFW (0.0642)

Do not accept a fraction as answer

*A **correct** answer can imply 4 marks*

A1

Special Case:

(Any given subject total) + 160 seen anywhere in (b)

Can award if no marks scored in (b)

Accept a decimal equivalent

(M1)

4

[11]

Q22.

(a) (i) $P(D \cap A) = 0.85 \times 0.80$

Can be implied by correct answer

M1

$$= 0.68$$

CAO; OE

A1

2

(ii) $P(D' \cap A) = 0.15 \times 0.30$

Can be implied by correct answer

M1

$$= 0.045$$

CAO; OE

A1

2

(iii) $P(\text{One}) = (ii) + (0.85 \times 0.20)$

Can be implied by correct answer

M1

$$= 0.045 + 0.17 = 0.215$$

CAO; OE

A1

2

(b) $P(D' \cap A') =$
 0.15×0.7

or

$$= 0.105$$

CAO; OE; must be stated somewhere

B1

$$1 - (0.68 + 0.215)$$

$$P((D \cap A)^3 \cap (D' \cap A')) =$$

$$[(a)(i)]^3 \text{ or } 0.68^3$$

M1

$$\times 0.105 \times 4$$

Either; with ≥ 1 ($\times$ sign)

M1

$$= 0.132$$

AWRT (0.13206)

A1

4

[10]

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Q23.

(a) (i) $P(G') = 1 - 0.70 = 0.3(0)$

CAO; OE

B1

1

(ii) $P(G \cap S') = 0.70 - (0.25 \text{ or } 0.55 \text{ or } 0.45)$
 or $1 - 0.55$

Can be implied only if answer is correct

M1

$$= 0.45$$

CAO; OE

A1

2

(iii) $P(1 \text{ only}) = 0.70 + 0.55 - (2 \times 0.25)$
 or $1 - 0.25$
 or $0.45 + 0.30$

*Can be implied only if answer is correct;
 allow no ($\times 2$) but not by implication from answer*

M1

$= 0.75$

CAO; OE

A1

2

(b) $P(G' \cap G' \cap G' \cap G') = [(a)(i)]^4$

*Can be implied by correct answer
 Ignore multiplier of 4*

M1

$= 0.0081$

CAO; OE

A1

2

(c) $P(H_G) = P(A_G \cap H_G) + P(A_{G'} \cap H_G) =$
 $(0.70 \times 0.60) \text{ or } 0.42$

*Can be implied by correct answer
 Ignore additional terms*

M1

$(0.30 \times 0.10) \text{ or } 0.03$

Can be implied by correct answer

M1

$= 0.42 + 0.03 = 0.45$

CAO; OE

A1

3

(d) $P(H_O) = 1 - [0.35 + (c)]$

Can be implied by correct answer

M1

$= 0.2(0)$

CAO; OE

A1

2

[12]

Q24.

- (a) Use of binomial in (a) or (b)
Can be implied by answers

M1

(i) $P(X \geq x) = 1 - P(X \leq x - 1)$

OR

$= 1 - B(\Sigma x, 50, 0.08)$

Identified from an answer / 1 – answer

Can be implied from a correct answer

Identified from an answer/expression

M1

$= 1 - 0.0827 = 0.915 \text{ to } 0.92(0)$

AWFW (0.9173)

A1

$\geq 1 \text{ correct} \Rightarrow \text{M1 M1}$

(ii) $P(X \geq 3)$
 $= 1 - 0.2260 = 0.77(0) \text{ to } 0.775$

AWFW (0.7740)

A1

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4

- (b) (i) $P(Y = 0) = (1 - 0.025)^{15} = 0.975^{15}$
Can be implied from correct answer

M1

$= 0.68(0) \text{ to } 0.685$

AWFW (0.6840)

A1

- (ii) $P(Y \geq 1) = 1 - (i)$
Can be implied from answer if $\epsilon (0, 1)$

M1

$$= 0.315 \text{ to } 0.32(0)$$

ft on (i) if $\varepsilon (0, 1)$ (0.3160)

A1ft

4

(c) Probability = [(b)(ii) $\times$ (a)(i)] or (0.316×0.917)

M1

$$[(b)(i) \times (a)(ii)] \text{ or } (0.684 \times 0.774)$$

M1

Ignore additional terms

$$= 0.2898 + 0.529$$

2 terms added with ≥ 1 correct

A1

$$= 0.81 \text{ to } 0.83$$

AWFW (0.8193)

A1

4

[12]

Q25.

(a) $P(R' \cap S' \cap T) = 0.3 \times 0.4 \times 0.8$

At least 1 probability correct

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M1

$$= 0.096$$

CAO; OE

A1

2

(b) $P(\text{One}) = (a) + P(R \cap S' \cap T') + P(R' \cap S' \cap T')$

Use of 3 possibilities, ignore multipliers

M1

$$= (a) + (0.7 \times 0.4 \times 0.2) + (0.3 \times 0.6 \times 0.2)$$

At least 1 new term correct

M1

$$= 0.096 + 0.056 + 0.036 = 0.188$$

CAO; OE

A1

3

- (c) $P(\text{At least one}) = 1 - P(\text{None})$
or $= P(\text{One or two or three})$
Used; OE; ignore multipliers

M1

$$= 1 - (0.3 \times 0.4 \times 0.2)$$

$$= 1 - 0.024$$

At least 1 new term correct

M1

or

$$= (b) + (0.7 \times 0.6 \times 0.2) + (0.7 \times 0.4 \times 0.8)$$

$$+ (0.3 \times 0.6 \times 0.8) + (0.7 \times 0.6 \times 0.8)$$

$$= 0.188 + 0.084 + 0.224 + 0.144 + 0.336$$

$$= 0.976$$

CAO; OE

A1

3

[8]

Q26.

- (a) $P(D' \cap E' \cap F') = 0.4 \times 0.3 \times 0.2$
At least 1 probability correct

M1

$$= 0.024$$

CAO; OE

A1

2

- (b) $P(D' \cap E' \cap F) = 0.4 \times 0.3 \times 0.8$
At least 2 probabilities correct

M1

$$= 0.096$$

CAO; OE

A1

2

- (c) $P(\text{One}) = (b) + P(D \cap E' \cap F') + P(D' \cap E \cap F')$

Use of 3 possibilities; ignore multipliers

M1

$$= (b) + (0.6 \times 0.3 \times 0.2) + (0.4 \times 0.7 \times 0.2)$$

At least 1 new term correct

M1

$$= 0.096 + 0.036 + 0.056 = 0.188$$

CAO; OE

A1

3

- (d) P(One or two)
 = (c) + (3 terms each of 3 probabilities)
or
 = $1 - (a) - (1 \text{ term of 3 probabilities})$
 (c) + P(Two)
 Used; OE; ignore multipliers
 $1 - (a) - P(\text{Three})$

M1

$$\begin{aligned}
 &= 0.188 + (0.6 \times 0.7 \times 0.2) + \\
 &\quad (0.6 \times 0.3 \times 0.8) + (0.4 \times 0.7 \times 0.8) \\
 &= 0.188 + 0.084 + 0.144 + 0.224
 \end{aligned}$$

At least 1 new term correct

M1

or
 $= 1 - 0.024 - (0.6 \times 0.7 \times 0.8)$
 $= 1 - 0.024 - 0.336$

$$= 0.64$$

CAO; OE

A1

3

[10]

Q27.

(a) $P(X) = 0.3 \quad P(Y) = 0.4 \quad P(Z) = 0.2$

(i) $P(X \cap Y \cap Z) = 0.3 \times 0.4 \times 0.2 = 0.024$

M1

1

(ii) $P(X' \cap Y' \cap Z') = 0.7 \times 0.6 \times 0.8$

At least 2 correct terms

M1

$$= 0.0336$$

CAO

A1

2

(iii) $P(X' \cap Y' \cap Z) = 0.7 \times 0.6 \times 0.2$

Correct numerical expression

M1

$$= 0.084$$

CAO

A1

2

(b) $P(W | Z) = 0.9 \quad P(W | Z') = 0.25$

(i) $P(Z \cap W) = 0.2 \times 0.9$

Correct numerical expression

M1

$$= 0.18$$

CAO

A1

2

(ii) $P((Z \cap W) \cup (Z' \cap W))$

or

$$1 - [P((Z \cap W) \cup (Z' \cap W'))]$$

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$$= 0.2 \times (1 - 0.9)$$

M1

$$+$$

$$0.2 \times 0.9 \quad \text{or} \quad (b) (i)$$

$$(1 - 0.2) \times 0.25$$

$$(1 - 0.2) \times (1 - 0.25)$$

Cannot score an M1 in both methods

M1

$$= 0.02 + 0.20$$

$$1 - (0.18 + 0.60)$$

$$= 0.22$$

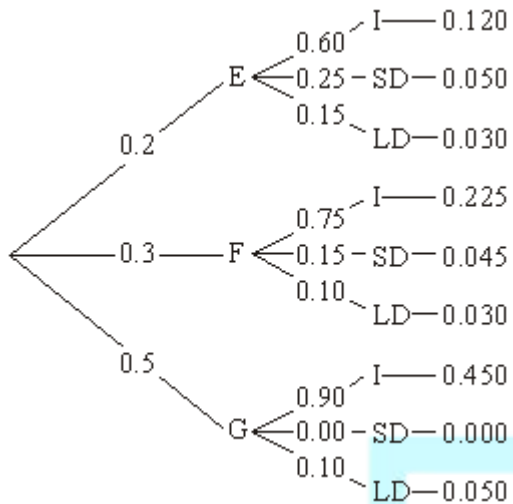
CAO

A1

3

[10]

Q28.



(a) (i) $P(G \cap I) = 0.5 \times 0.9 = 0.45$
CAO; or equivalent

B1

1

(ii) $P(I) = (i) + P(E \cap I) + P(F \cap I)$
3 possibilities

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M1

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3

$$= 0.45 + (0.2 \times 0.6) + (0.3 \times 0.75)$$

≥ 1 correct **new term**

A1

$$= 0.45 + 0.12 + 0.225 = 0.795$$

CAO; or equivalent

A1

3

(iii) $P(G | I) = \frac{P(G \cap I)}{P(I)}$

Attempted use of Bayes' Theorem

M1

$$= \frac{\frac{(i)}{(ii)}}{\frac{(i)}{(ii)}} = \frac{0.45}{0.795} = 0.566$$

m1

AWRT; or equivalent

A1

3

$$(b) \quad P(E | SD) = \frac{P\{E \cap SD\}}{P\{SD\}}$$

Correct use of Bayes' Theorem

M1

$$= \frac{0.2 \times 0.25}{(0.2 \times 0.25) + (0.3 \times 0.15)} =$$

Numerator (B1 if no Bayes' Theorem)

A1

$$\frac{0.05}{0.05 + 0.045}$$

Denominator (B1 if no Bayes' Theorem)

A1

$$= \frac{0.05}{0.095} = 0.526$$

AWRT; or equivalent

A1

4

Q29.

$$\underline{P(D) = 0.6 \text{ (60\%)}} \\ \underline{(40\%)}$$

$$\underline{P(d | D) = 0.75}$$

$$\underline{P(i | D) = 0.15}$$

$$\underline{P(n | D) = 0.10}$$

$$\underline{P(D') = 0.4}$$

$$\underline{P(d | D') = 0.05}$$

$$\underline{P(i | D') = 0.15}$$

$$\underline{P(n | D') = 0.80}$$

$D \Rightarrow$ Diseased fish

$d \Rightarrow$ diseased

$i \Rightarrow$ inconclusive test

$n \Rightarrow$ non-diseased

(a) (i) $P(D \cap d)$
 $= P(D) \times P(d | D) = 0.6 \times 0.75$

M1

$$= 0.45$$

CAO (9/20)

A1

2

(ii) $P(d) = P(D \cap d) + P(D' \cap d)$
 $= (i) + P(D') \times P(d | D')$
 (a)(i) + (1 prob)

M1

$$= 0.45 + (1 - 0.6) \times 0.05$$

ft on (a)(i)

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m1ft

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$$= 0.45 + 0.02 = 0.47$$

CAO

A1

3

(iii) $P(\text{correct}) = P(D \cap d) + P(D' \cap d')$
 $= (i) + P(D') \times P(d' | D')$

Correct expression/values

ft on (a)(i)

M1ft

Or

$$= 0.45 + (1 - 0.6) \times 0.80$$

$$= 0.45 + 0.32 = 0.77$$

CAO

A1

2

$$\begin{aligned}
 (b) \quad & P([d | D] \cap [d | D] \cap [d' | D]) \\
 &= [P(d | D)]^2 \times P(d' | D) \\
 &\quad (Prob)^2 \times (Prob); \text{ or equivalent}
 \end{aligned}$$

M1

multiplied by 3

Multiplier of 3

M1

$$\begin{aligned}
 &= 0.75^2 \times 0.10 \times 3 = 0.168 \text{ to } 0.169 \\
 &\quad \text{AWFW } (0.16875)
 \end{aligned}$$

A1

3

[10]

Q30.

$$\underline{P(F) = 0.8 \quad P(D | F) = 0.9 \quad P(D | F') = 0.4}$$

$$\begin{aligned}
 (a) \quad (i) \quad & P(F \cap D) \\
 &= P(F) \times P(D | F) = 0.8 \times 0.9
 \end{aligned}$$

M1

$$= 0.72$$

CAO (18/25)

A1

2

$$\begin{aligned}
 (ii) \quad & P(F' \cap D') \\
 &= P(F') \times P(D' | F') = (1 - 0.8) \times (1 - 0.4)
 \end{aligned}$$

M1

$$= 0.2 \times 0.6 = 0.12$$

CAO (3/25)

A1

2

$$(b) \quad \underline{P(M) = 0.7}$$

$$\begin{aligned}
 (i) \quad & P(F \cap D \cap M) = P(F) \times P(D | F) \times P(M) \\
 &\quad (a)(i) \times P(M), \text{ ignore multipliers}
 \end{aligned}$$

M1

$$= (a)(i) \times P(M) = 0.72 \times 0.7$$

Or equivalent; ft on (a)(i) < 1

A1ft

$$= 0.504$$

CAO (63/125)

A1

3

(ii) $P(2 \text{ in } 3)$

$$\begin{aligned}
 &= P(F \cap D \cap M') \\
 &+ P(F \cap D' \cap M) \\
 &+ P(F' \cap D \cap M)
 \end{aligned}$$

*At least 2 permutations of 3 events seen, or implied by
 multiplication of 3 correct probabilities at least twice
 Ignore multipliers e.g. $\times 3$*

M1

$$\begin{aligned}
 &= 0.8 \times 0.9 \times 0.3 \\
 &+ 0.8 \times 0.1 \times 0.7
 \end{aligned}$$

A2

$$+ 0.2 \times 0.4 \times 0.7$$

*At least 2 correct expressions
 (Exactly 1 correct expression)*

(A1)

$$\begin{aligned}
 &= 0.216 + 0.056 + 0.056 \\
 &= 0.328
 \end{aligned}$$

CAO (41/125)

A1

4

[11]

Q31.

(a) (i) $P(S1) = 0.3$

B1

1

(ii) $P(S1 \text{ \& } A) = 0.3 \times 0.2 = 0.06$

ft wrong answer to (i)

B1F

1

- (iii) $P(SI' \& A) = 0.7 \times 0.4$
M1 for reasonable attempt

M1

$$P(A) = 0.06 + 0.28 = 0.34$$

convincingly shown

A1

2

- (b) Cond prob with N or D correct
where $0 < N < D < 1$

M1

Numerator 0.06
ft wrong answer to (a)(ii)

A1F

Denominator 0.34

A1

3

[7]

Q32.

- (a) (i) $P(\text{both}) = 0.1 \times 0.05 = 0.005$

M1A

2

- (ii) $P(\text{neither}) = 0.9 \times 0.95 = 0.855$

M1A1

2

- (iii) $P(\text{exactly one}) = 0.14$
ft wrong values if subtraction from 1 used

M1A1F

2

- (b) Formula for conditional prob
*Fraction with $0 < N < D < 1$ and D correct
 or equal to c's answer to (a)(iii)*

M1

Numerator = 0.1×0.95

m1

Denom = 0.14 so ans = $\frac{19}{28}$

*ft wrong answer to (a)(iii);
Accept AWR 0.679 or 0.678*

A1F

3

[9]

Q33.

(a) (i) $P(\text{British}) = \frac{13}{30}$

Allow decimal answers throughout question

B1

1

(ii) $P(\text{goalkeeper}) = \frac{4}{30} \left(\frac{2}{15} \right)$

B1

1

(iii) $P(\text{British \& goalkeeper}) = \frac{2}{30} \left(= \frac{1}{15} \right)$

B1

1

(iv) $P(\text{defender or midfielder}) = \frac{21}{30} \left(= \frac{7}{10} \right)$

B1

1

(v) $P(\text{Br or attacker, not both}) = \frac{14}{30} \left(= \frac{7}{15} \right)$

*M1 for **fully** correct method*

M1A1

2

(b) (i) $P(\text{goalkeeper} | \text{British}) = \frac{2}{13}$

M1 for correct denominator

M1A1

2

(ii) Not independent (reason)

E1 if incompletely explained eg no numerical value

E2,1

2

[10]

Q34.

(a) (i) $P(\text{work \& TMR}) = \frac{2}{3} \times \frac{2}{5} = \frac{4}{15}$
'TMR' means 'ten mile run'

M1A1

2

(ii) $P(\text{no work, TMR}) = \frac{1}{3} \times \frac{4}{5} = \frac{4}{15}$

M1A1

2

(iii) $P(\text{TMR}) = \frac{4}{15} + \frac{4}{15} = \frac{8}{15}$
ft errors in (i) and /or (ii)

A1F

(b) Reasonable attempt

$\frac{x}{y}$
 Allow $\frac{x}{y}$ where $0 < x < y = (a)(iii)$

M1

$P(\text{work} | \text{TMR}) = \frac{1}{2}$

Allow NMS; ft errors in (a)

A1F

2

[7]

Q35.

(a) (i) $P(\text{both}) = 0.9 \times 0.3 = 0.27$

M1A1

2

(ii) $P(\text{neither}) = 0.1 \times 0.7 = 0.07$

M1A1

2

- (iii) $P(\text{at least one}) = 1 - 0.07 = 0.93$
ft wrong answer to (ii)

M1A1F

2

- (b) Conditional probability = $\frac{0.9}{0.93} \left(= \frac{90}{93} \right)$
ft wrong answer to (a)(iii) provided it lies between 0.9 and 1
Condone failure to simplify $\frac{0.9}{0.93}$
NMS 2/3 for AWRT 0.97

B3F

3

[9]



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